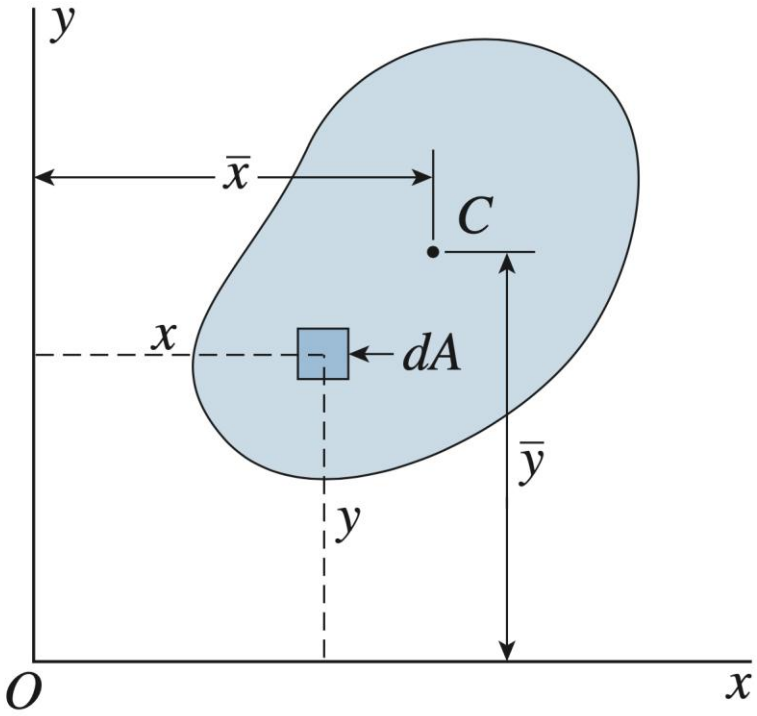
Bending Stress

Note: content marked in * is more advanced knowledge

First moment of an area



The **area** of the geometric figure is defined by the following integral:

$$A = \int dA$$

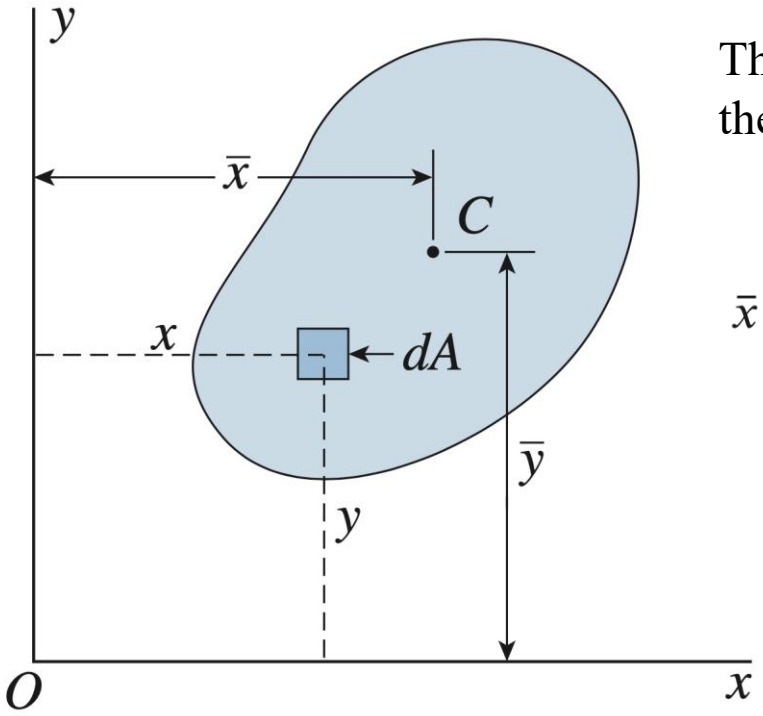
In which dA is a differential element of area having coordinates x and y

The **first moments** of the area with respect to the x and y axes are defined, respectively, as follows:

$$Q_x = \int y \, dA \qquad Q_y = \int x \, dA$$

- The first moments represent the sums of the products of the differential areas and their coordinates
- First moments may be positive or negative, depending upon the position of the xy axes

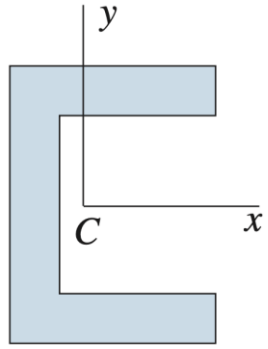
First moment of an area

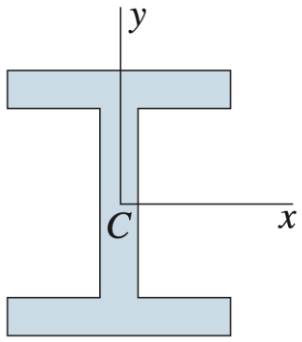


The coordinates $\bar{x}$ and $\bar{y}$ of the **centroid** C are equal to the first moments divided by the area:

$$\bar{x} = \frac{Q_y}{A} = \frac{\int x\ dA}{\int dA} \qquad \bar{y} = \frac{Q_x}{A} = \frac{\int y\ dA}{\int dA}$$

$$Q_x = A\bar{y} \qquad Q_y = A\bar{x}$$

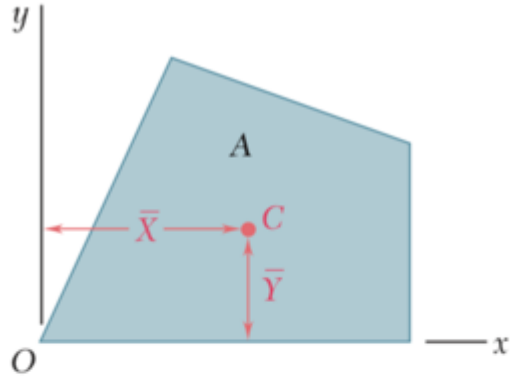




- If an area is **symmetric about an axis**, the centroid must lie on that axis because the first moment about an axis of symmetry equals zero
- If an area has **two axes of symmetry**, the position of the centroid can be determined by inspection because it lies at the **intersection** of the axes of symmetry.

The centroid of a composite area

The **areas** and **first moments** of composite areas may be calculated by summing the corresponding properties of the component parts.

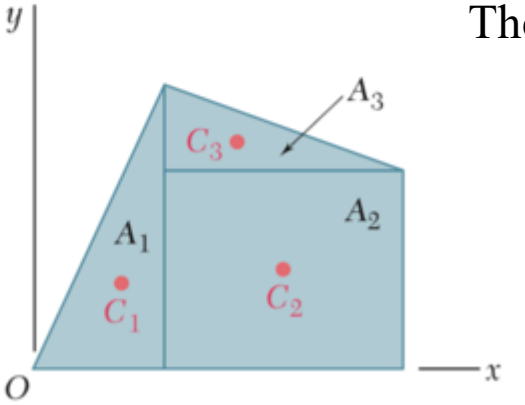


An area A can be divided into **simple** geometric shapes

$$Q_x = \int_A y\,dA = \int_{A_1} y\,dA + \int_{A_2} y\,dA + \int_{A_3} y\,dA$$

$$Q_x = A_1\bar{y}_1 + A_2\bar{y}_2 + A_3\bar{y}_3$$

=



The first moment of the area A can also be expressed as:

$$Q_x = A\bar{Y} = \sum_i A_i\bar{y}_i$$

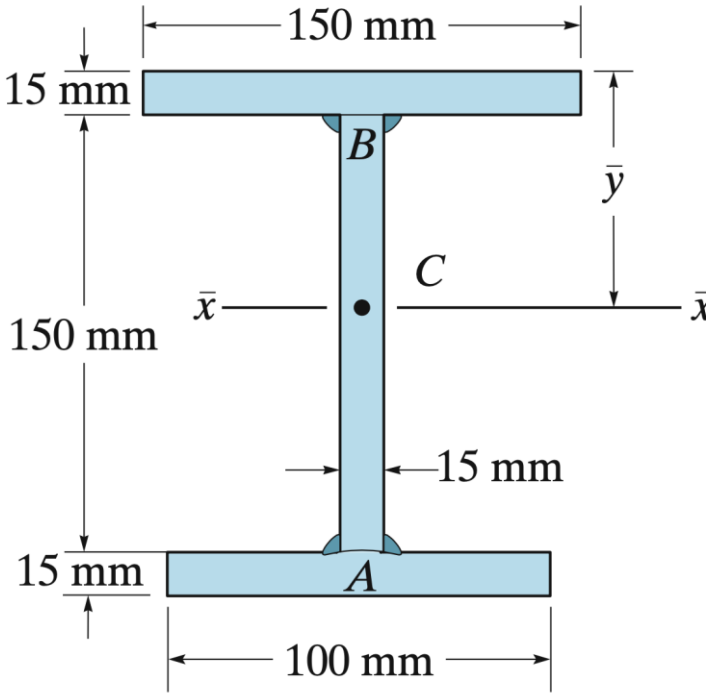
$$\bar{X} = \frac{\sum_i A_i\bar{x}_i}{\sum_i A_i} \qquad \bar{Y} = \frac{\sum_i A_i\bar{y}_i}{\sum_i A_i}$$

Example 1: centroid of a composite area

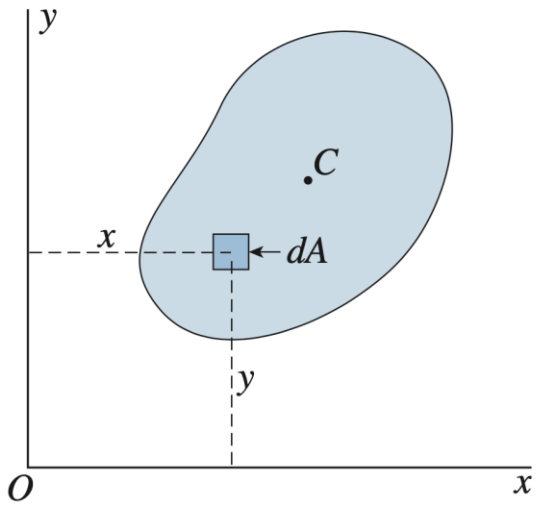
Question : Locate the centroid C of the cross-section

$$\bar{y} = \frac{\Sigma \tilde{y} A}{\Sigma A} = \frac{7.5(15)(150) + 90(150)(15) + 172.5(15)(100)}{15(150) + 150(15) + 15(100)}$$

$$= 79.6875 \text{ mm} = 79.7 \text{ mm}$$



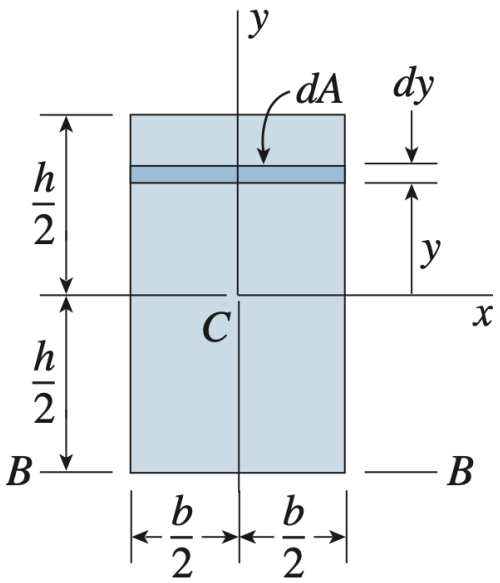
Second moment of an area (Moment of inertia)



The **second moment of the area** (also called **moments of inertia**) A with respect to the x (or y) axis:

$$I_x = \int y^2\,dA \qquad I_y = \int x^2\,dA$$

Example 2: Determine the moment of inertia I_x of the area with respect to the centroidal x axis:



$$dI_x = y^2\,dA = y^2(b\,dy)$$

Integrating from $y = -h/2$ to $y = +h/2$, we write

$$I_x = \int_A y^2\,dA = \int_{-h/2}^{+h/2} y^2(b\,dy) = \tfrac{1}{3}b[y^3]_{-h/2}^{+h/2}$$

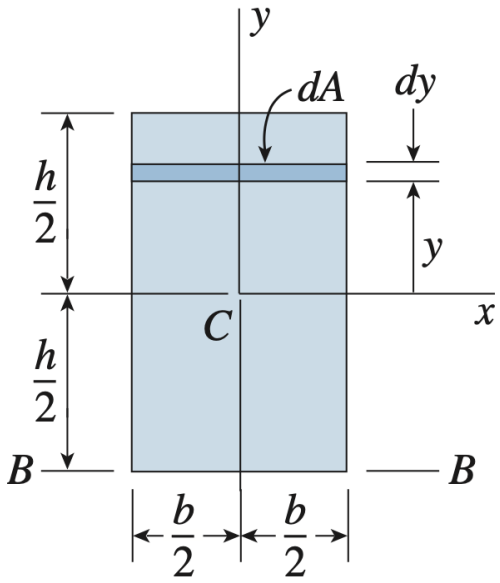
$$= \tfrac{1}{3}b\left(\frac{h^3}{8} + \frac{h^3}{8}\right) = \frac{bh^3}{12}$$

Second moment of an area (Moment of inertia)

If a different set of axes is selected, the moments of inertia will have different values

If axis *BB* at the base of the rectangle is selected as the reference, we must define *y* as the coordinate distance from that axis to the element of area *dA*.
Then the calculations for the moment of inertia become

$$I_{BB} = \int y^2 dA = \int_0^h y^2 b\, dy = \frac{bh^3}{3}$$

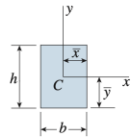


Note: the moment of inertia with respect to axis *BB* is larger than the moment of inertia with respect to the centroidal *x* axis

In general, the moment of inertia increases as the reference axis is moved parallel to itself farther from the centroid.

Second moment of an area: Expression is provided

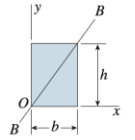
1 Rectangle (Origin of axes at centroid)



$$A = bh \quad \bar{x} = \frac{b}{2} \quad \bar{y} = \frac{h}{2}$$

$$I_x = \frac{bh^3}{12} \quad I_y = \frac{hb^3}{12} \quad I_{xy} = 0 \quad I_P = \frac{bh}{12}(h^2 + b^2)$$

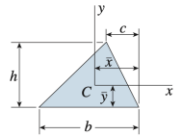
2 Rectangle (Origin of axes at corner)



$$I_x = \frac{bh^3}{3} \quad I_y = \frac{hb^3}{3} \quad I_{xy} = \frac{b^2h^2}{4} \quad I_P \frac{bh}{3}(h^2 + b^2)$$

$$I_{BB} = \frac{b^3h^3}{6(b^2 + h^2)}$$

3 Triangle (Origin of axes at centroid)

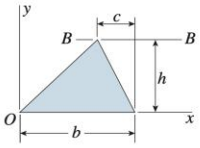


$$A = \frac{bh}{2} \quad \bar{x} = \frac{b + c}{3} \quad \bar{y} = \frac{h}{3}$$

$$I_x = \frac{bh^3}{36} \quad I_y = \frac{bh}{36}(b^2 - bc + c^2)$$

$$I_{xy} = \frac{bh^2}{72}(b - 2c) \quad I_P = \frac{bh}{36}(h^2 + b^2 - bc + c^2)$$

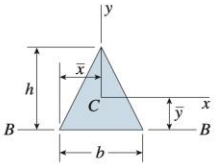
4 Triangle (Origin of axes at vertex)



$$I_x = \frac{bh^3}{12} \quad I_y = \frac{bh}{12}(3b^2 - 3bc + c^2)$$

$$I_{xy} = \frac{bh^2}{24}(3b - 2c) \quad I_{BB} = \frac{bh^3}{4}$$

5 Isosceles triangle (Origin of axes at centroid)



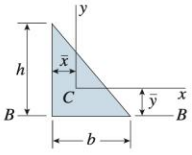
$$A = \frac{bh}{2} \quad \bar{x} = \frac{b}{2} \quad \bar{y} = \frac{h}{3}$$

$$I_x = \frac{bh^3}{36} \quad I_y = \frac{hb^3}{48} \quad I_{xy} = 0$$

$$I_P = \frac{bh}{144}(4h^2 + 3b^2) \quad I_{BB} = \frac{bh^3}{12}$$

(*Note:* For an equilateral triangle, $h = \sqrt{3}\, b/2$.)

6 Right triangle (Origin of axes at centroid)

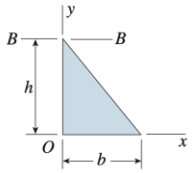


$$A = \frac{bh}{2} \quad \bar{x} = \frac{b}{3} \quad \bar{y} = \frac{h}{3}$$

$$I_x = \frac{bh^3}{36} \quad I_y = \frac{hb^3}{36} \quad I_{xy} = -\frac{b^2h^2}{72}$$

$$I_P = \frac{bh}{36}(h^2 + b^2) \quad I_{BB} = \frac{bh^3}{12}$$

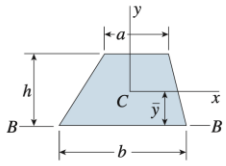
7 Right triangle (Origin of axes at vertex)



$$I_x = \frac{bh^3}{12} \quad I_y = \frac{hb^3}{12} \quad I_{xy} = \frac{b^2h^2}{24}$$

$$I_P = \frac{bh}{12}(h^2 + b^2) \quad I_{BB} = \frac{bh^3}{4}$$

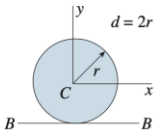
8 Trapezoid (Origin of axes at centroid)



$$A = \frac{h(a + b)}{2} \quad \bar{y} = \frac{h(2a + b)}{3(a + b)}$$

$$I_x = \frac{h^3(a^2 + 4ab + b^2)}{36(a + b)} \quad I_{BB} = \frac{h^3(3a + b)}{12}$$

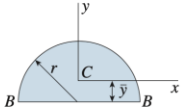
9 Circle (Origin of axes at center)



$$A = \pi r^2 = \frac{\pi d^2}{4} \quad I_x = I_y = \frac{\pi r^4}{4} = \frac{\pi d^4}{64}$$

$$I_{xy} = 0 \quad I_P = \frac{\pi r^4}{2} = \frac{\pi d^4}{32} \quad I_{BB} = \frac{5\pi r^4}{4} = \frac{5\pi d^4}{64}$$

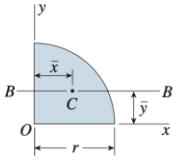
10 Semicircle (Origin of axes at centroid)



$$A = \frac{\pi r^2}{2} \quad \bar{y} = \frac{4r}{3\pi}$$

$$I_x = \frac{(9\pi^2 - 64)r^4}{72\pi} \approx 0.1098r^4 \quad I_y = \frac{\pi r^4}{8} \quad I_{xy} = 0 \quad I_{BB} = \frac{\pi r^4}{8}$$

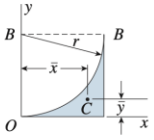
11 Quarter circle (Origin of axes at center of circle)



$$A = \frac{\pi r^2}{4} \quad \bar{x} = \bar{y} = \frac{4r}{3\pi}$$

$$I_x = I_y = \frac{\pi r^4}{16} \quad I_{xy} = \frac{r^4}{8} \quad I_{BB} = \frac{(9\pi^2 - 64)r^4}{144\pi} \approx 0.05488r^4$$

12 Quarter-circular spandrel (Origin of axes at point of tangency)



$$A = \left(1 - \frac{\pi}{4}\right)r^2 \quad \bar{x} = \frac{2r}{3(4 - \pi)} \approx 0.7766r \quad \bar{y} = \frac{(10 - 3\pi)r}{3(4 - \pi)} \approx 0.2234r$$

$$I_x = \left(1 - \frac{5\pi}{16}\right)r^4 \approx 0.01825r^4 \quad I_y = I_{BB} = \left(\frac{1}{3} - \frac{\pi}{16}\right)r^4 \approx 0.1370r^4$$

Extracted from Appendix D of Mechanics of Materials, Eighth Edition, SI by James M. Gere and Barry J. Goodno

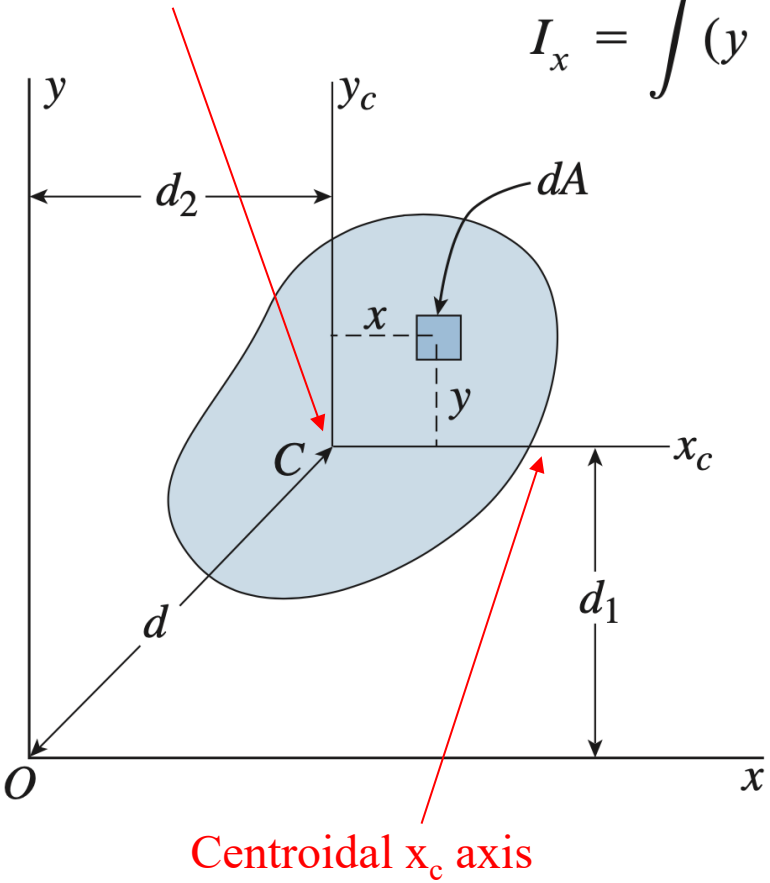
Second moment of an area: parallel-axis theorem

From the definition of moment of inertia, we can write the following equation for the moment of inertia I_x with respect to the x axis

$$I_x = \int (y + d_1)^2 dA = \int y^2 dA + 2d_1 \int y\, dA + d_1^2 \int dA$$

$$I_x = I_{x_c} + Ad_1^2$$

It equals zero because the x_c axis passes through the centroid



Parallel-axis theorem : *The moment of inertia of an area with respect to any axis in its plane is equal to the moment of inertia with respect to a parallel centroidal axis plus the product of the area and the square of the distance between the two axes.*

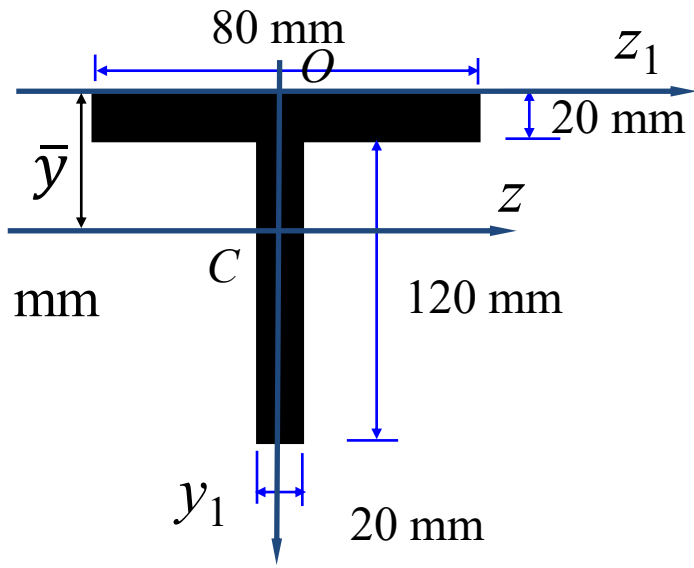
In previous example $I_{BB} = I_x + Ad^2 = \frac{bh^3}{12} + bh\left(\frac{h}{2}\right)^2 = \frac{bh^3}{3}$

Example 2: moment of inertia of a composite area

Question: Determine the moment of inertia *Ic* with respect to the horizontal *z* axis through the centroid *C* of the beam cross section

1. Centroid:

$$\bar{y} = \frac{\sum A_i y_i}{\sum A_i} = \frac{80\times 20\times 10 + 20\times 120\times (60+20)}{80\times 20 + 20\times 120} = 52 \text{ mm}$$

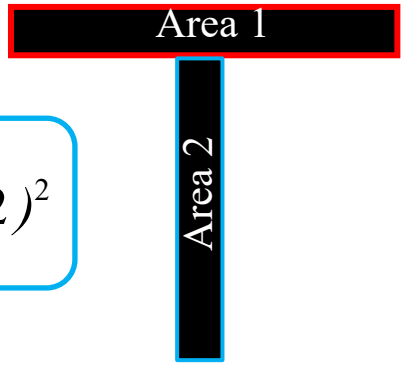


2. Moment of inertia:

By the Parallel Axis Theorem: $I_{z'} = I_z + Ad_{zz'}^2$

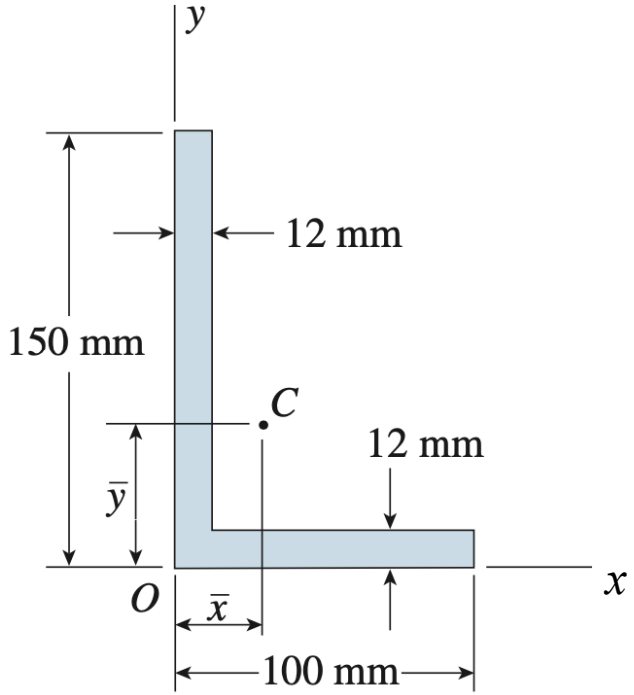
$$I_z = \underbrace{\frac{80\times 20^3}{12} + 80\times 20\times (52-10)^2}_{\text{Area 1}} + \underbrace{\frac{20\times 120^3}{12} + 20\times 120\times (80-52)^2}_{\text{Area 2}}$$

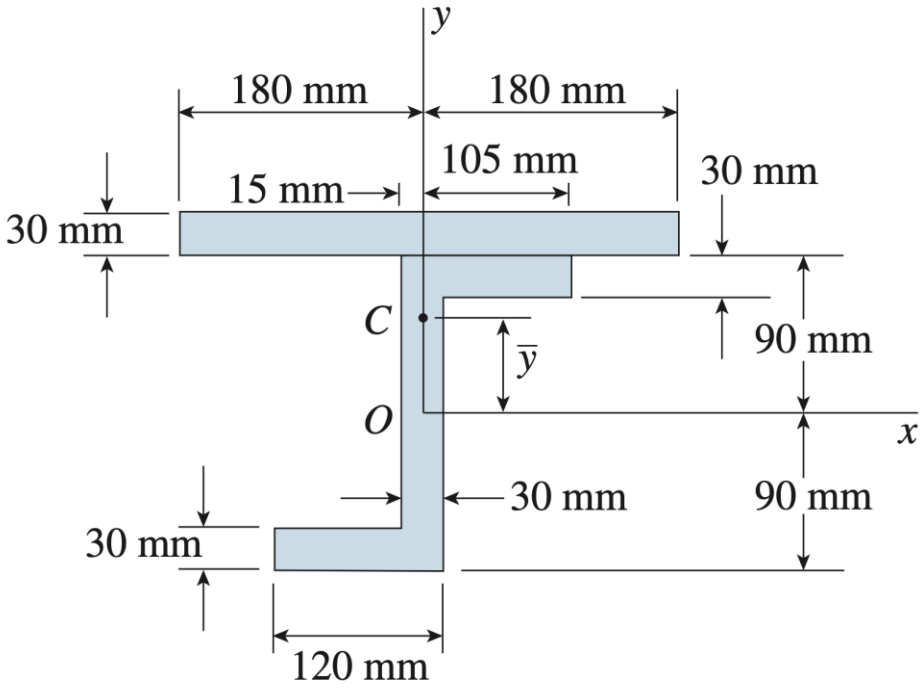
$$= 764\times 10^4 \text{ mm}^4 = 7.64\times 10^{-6} \text{ m}^4$$



Exercise

Question: Locate the centroid and calculate the moment of inertia with respect to an axis through the centroid C





Answers:

$\bar{x} = 24.5$ mm, $\bar{y} = 49.5$ mm

$I_x = 13.55 \times 10^6$ mm^4,

$I_y = 4.08 \times 10^6$ mm^4

$I_{xc} = 6.56 \times 10^6$ mm^4

(I_x and I_y are moments of inertia with respect to the axes x and y shown in the figures)

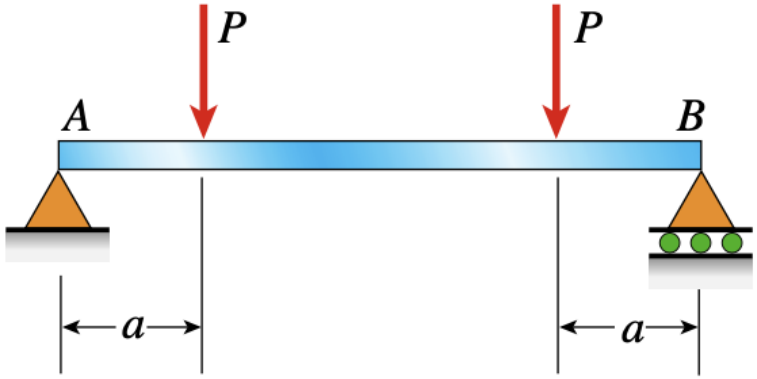
Answers:

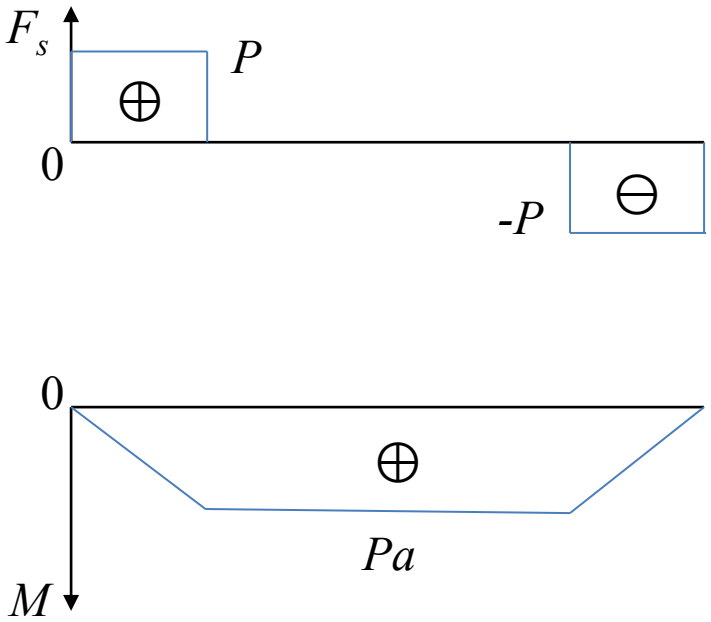
$\bar{y} = 52.5$ mm

$I_{x_c} = 106 \times 10^6$ mm^4

Pure Bending

Pure bending refers to the flexure of a beam under a constant bending moment (shear force is zero, because $F_s(x)=dM(x)/dx$).

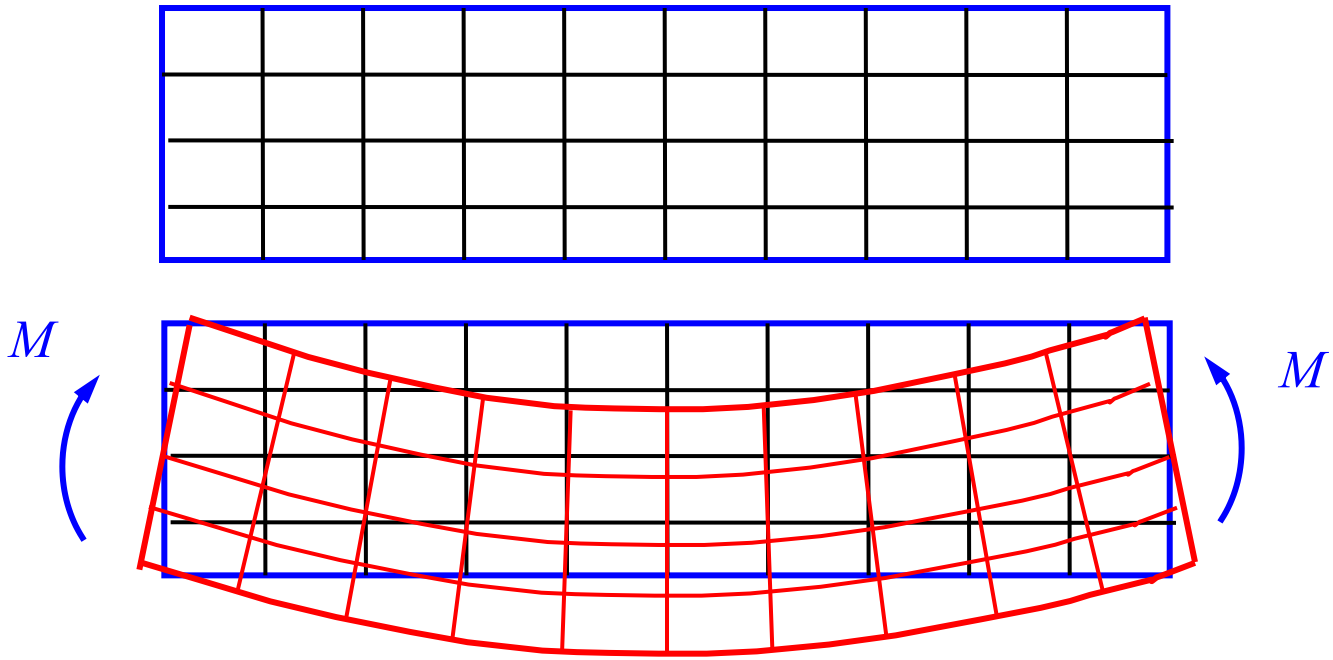




Deformation Characteristics

- Before:

- After:



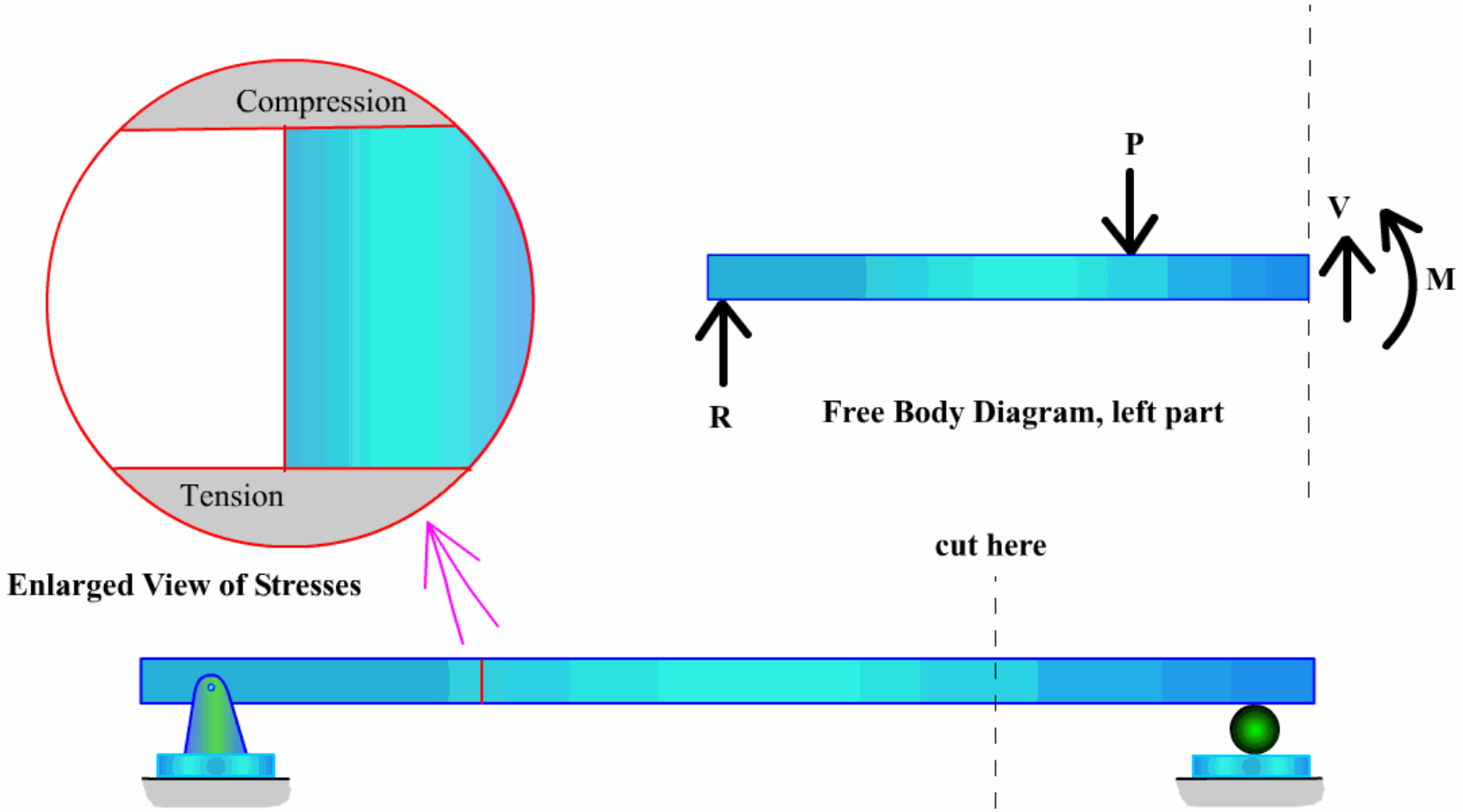
- Straight longitudinal lines turns into curves.
- Longitudinal lines get shortened under compression and elongated under tension.
- Cross-section lines remain straight and perpendicular to longitudinal curves (**plane assumption/theorem**).

Assumptions for Pure Bending

- **Plane assumption:** under pure bending, cross-sections of beams remain planar and perpendicular (normal) to beam axis and only rotate a small angle.

- **Assumption of uniaxial stress state:** individual longitudinal layers are under uniaxial tension/compression along beam axis, without stresses acting in between.

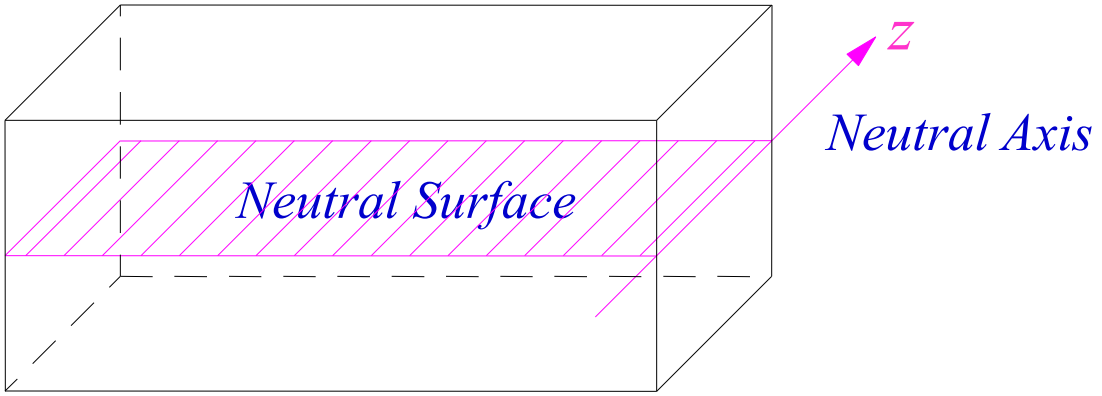
Bending Action



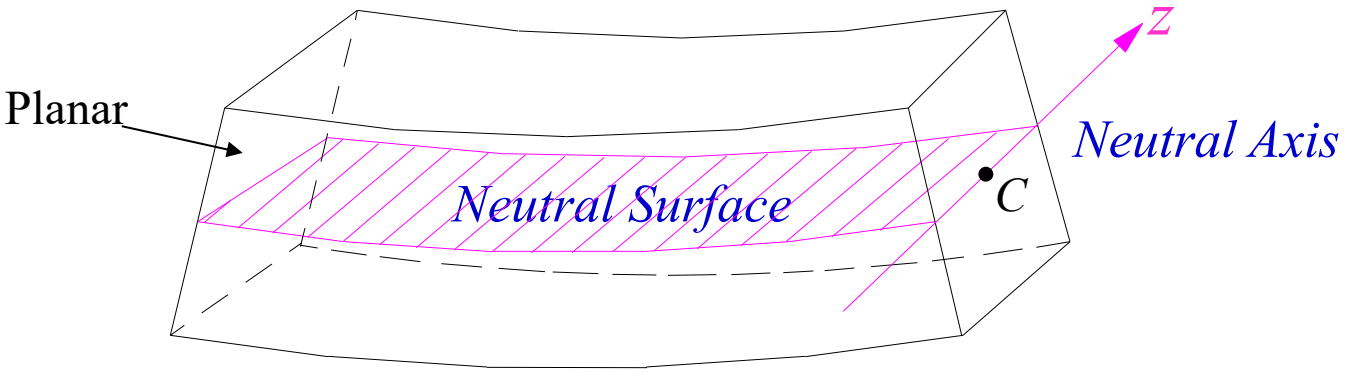
At cross-sections:
Internal Force/Moment = Resultant of Stress

Neutral Surface & Neutral Axes

- Before:

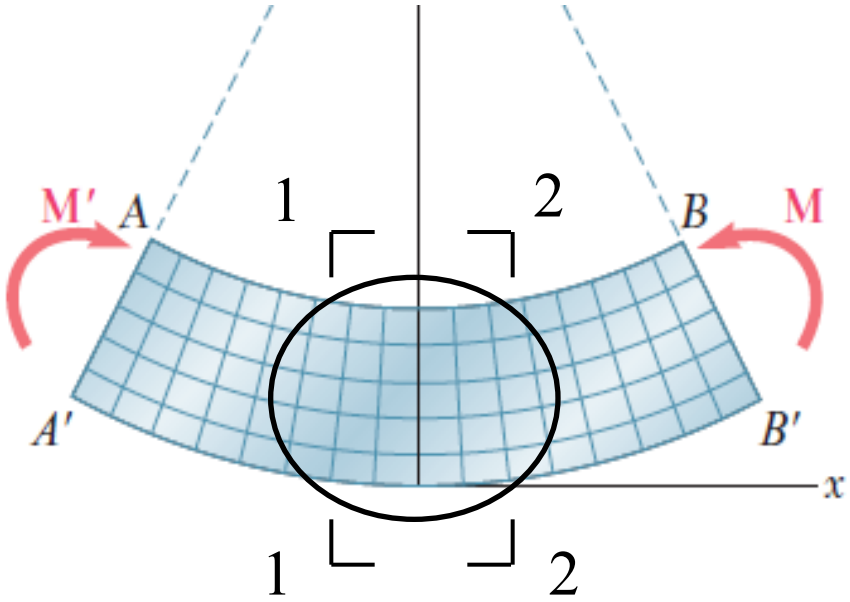


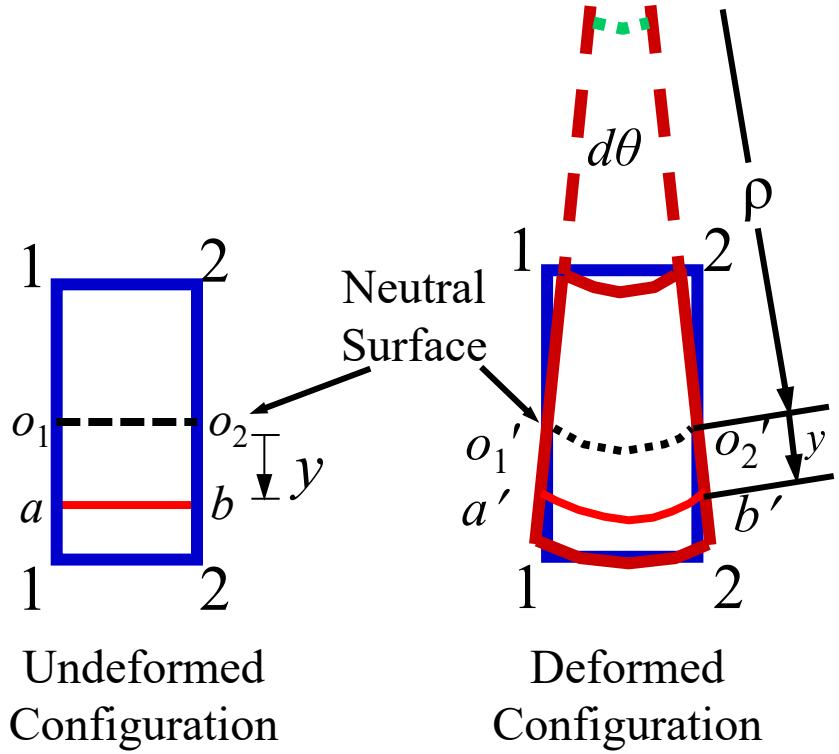
- After:



- **Neutral Surface :** the longitudinal layer under neither tension nor compression (its length remains unchanged during bending).
- **Neutral Axes:** intersecting lines of the neutral surface & cross sections.

Kinematics





Section 1-2

$$\varepsilon_{xx} = \varepsilon(y) = \frac{L_{a'b'} - L_{ab}}{L_{ab}}$$

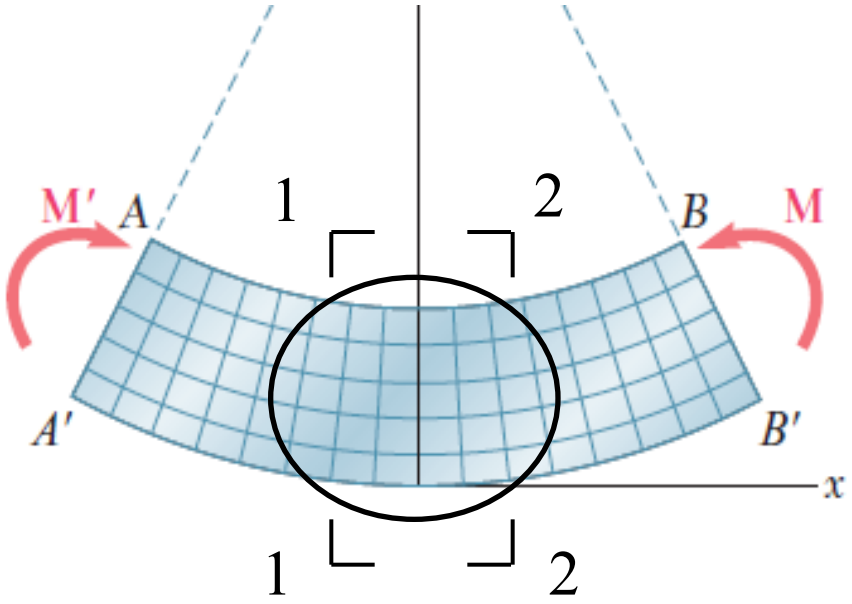
where $L_{a'b'} = (\rho + y)d\theta$,

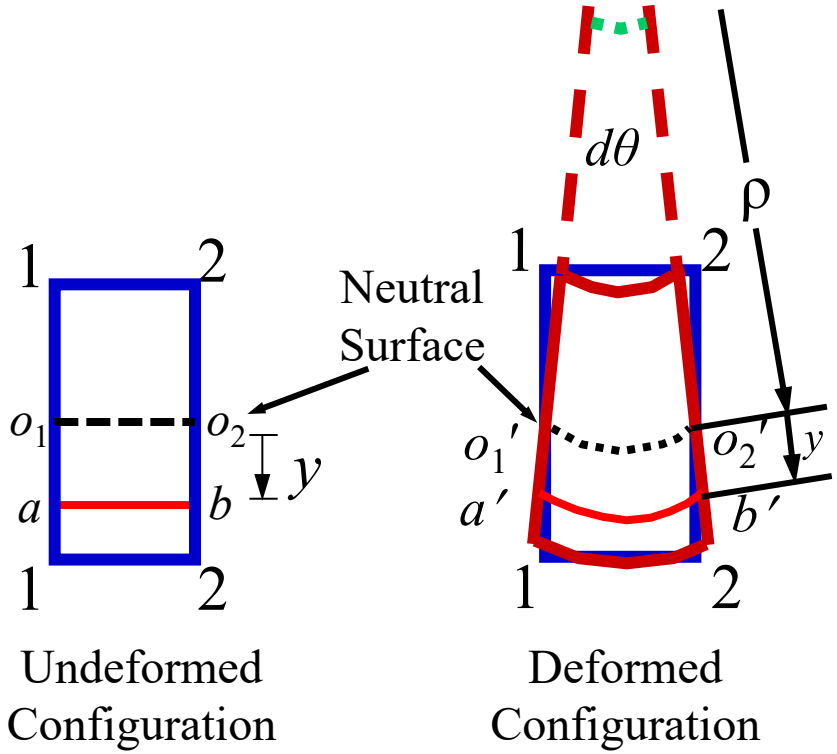
$L_{ab} = L_{o_1o_2} = L_{o_1'o_2'} = \rho d\theta$

(ρ is the radius of curvature)

- Normal stress σ (*y*) and strain ε (*y*) are *uniaxial* along *x*-coordinate.
- Length o_1-o_2 $L_{o_1o_2}$ is *constant* on the neutral surface (even after bending).

Kinematics



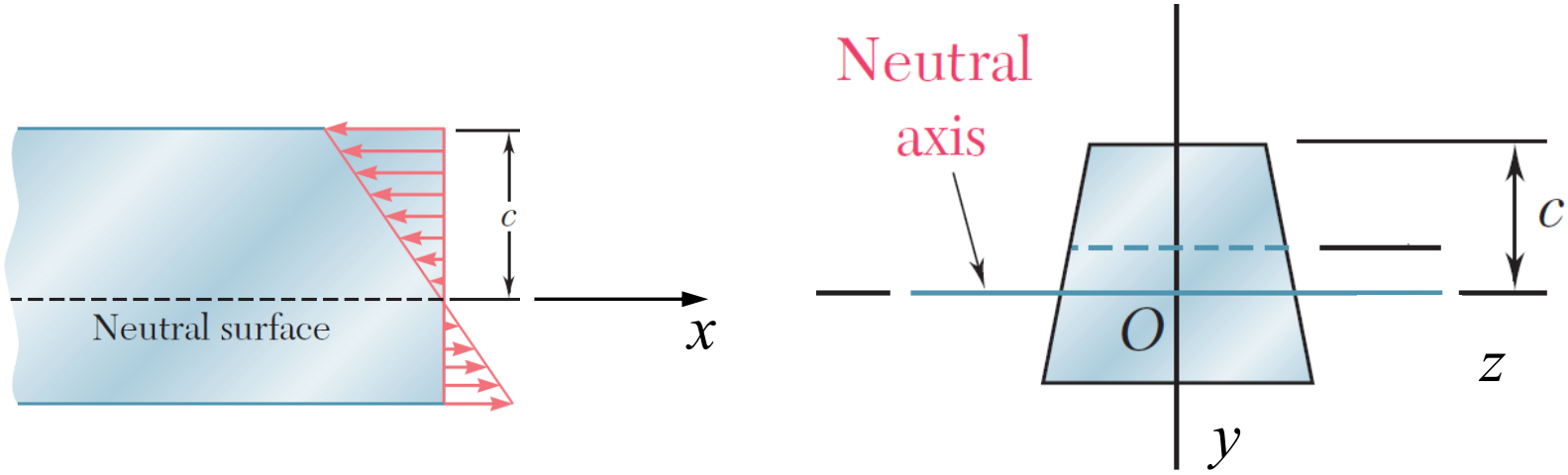


$$\varepsilon_{xx} = \varepsilon(y) = \frac{(\rho + y)d\theta - \rho d\theta}{\rho d\theta} = \frac{y}{\rho}$$

y : distance to the neutral axis

- The y-coordinate is measured from the proposed neutral axis.

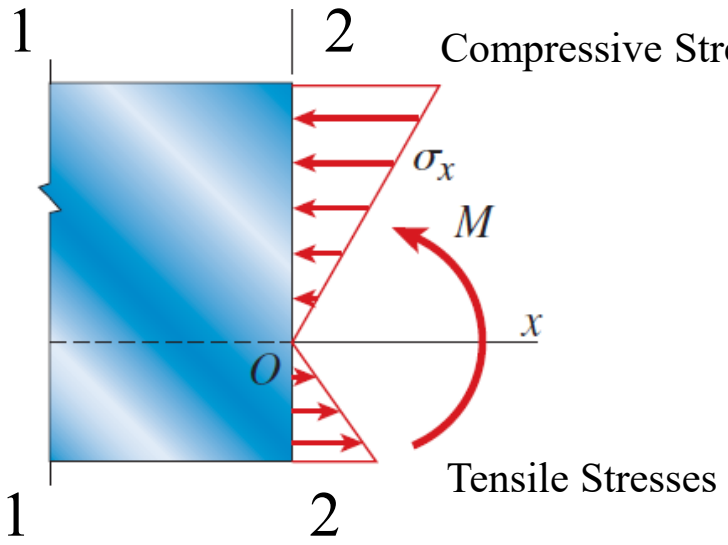
Hooke's Law



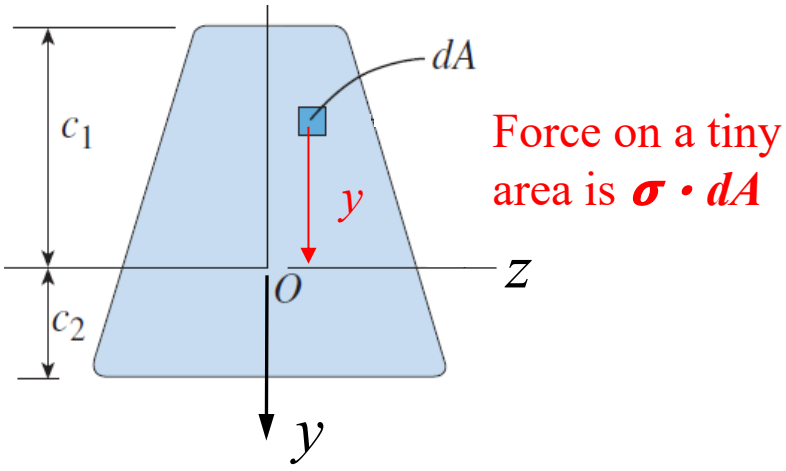
$$\sigma_{xx} = \sigma(y) = E\varepsilon(y) = E\frac{y}{\rho}$$

- Normal stress acting on a longitudinal layer is **linearly** proportional to its distance from the neutral surface, **positive for layers under tension / negative for layers under compression**.
- Remark: the above equation can only be used for qualitative analysis of stresses in bending beams since it is difficult to measure the curvature of radius (ρ) of individual longitudinal layers.

Static Equivalency (Internal Force = Resultants of stress)



Section 1-2



Cross Section

Force on a tiny area is $\boldsymbol{\sigma \cdot dA}$

- Static Equivalency (1): The resultant axial force, F_N, acting on the cross section is 0,

$$0 = F_N = \int_a \sigma dA = \int_a \frac{E}{\rho} y dA = \frac{E}{\rho}\int_a y dA \Rightarrow \int_a y dA = 0$$

First moment of the cross section with respect to z axis

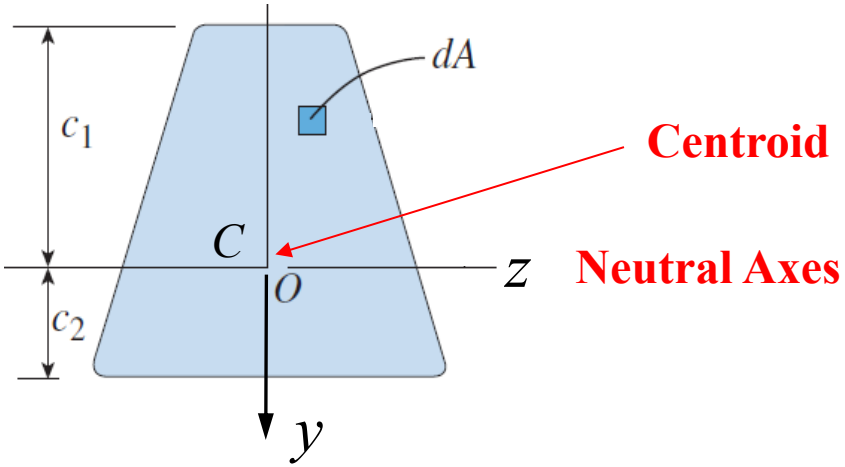
$$0 = \int_a y dA = A\bar{y} \Rightarrow \bar{y} = 0.$$

$\bar{y}$ is the coordinate of centroid of the cross section (distance to centroid)

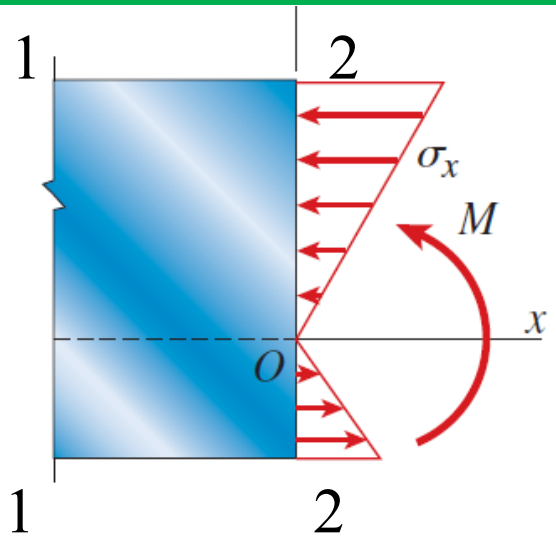
Static Equivalency

Important Conclusions:

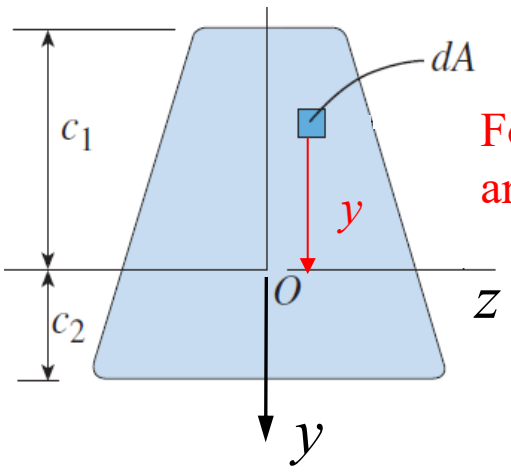
- The **neutral axis passes through the centroid** of the cross-sectional area when the material follows Hooke's law and there is no axial force acting on the cross section.
- Our discussion is limited to beams for which the *y* axis is an axis of symmetry (symmetric bending). Consequently, the *y* axis also passes through the centroid.
- Therefore, the **origin of coordinates is the centroid**.



Static Equivalency (Internal Force = Resultants of stress)



Section 1-2



Cross Section

Force on a tiny area is $\boldsymbol{\sigma \cdot dA}$

Static Equivalency (2): Moment resultant of the normal stresses over the entire cross section is equal to the bending moment

$$M_Z = \int_A y \cdot \sigma dA = \int_A \frac{E}{\rho} y^2 dA = \frac{E}{\rho} I_z \quad \Rightarrow \quad \boxed{\frac{1}{\rho} = \frac{M_z}{EI_z}}$$

Moment-curvature equation

$$\Rightarrow \sigma = \frac{E}{\rho} y \quad \Rightarrow \quad \boxed{\sigma = \frac{M_z y}{I_z}}$$

Flexure formula

EI_z : Flexural rigidity

$I_z = \int_A y^2 dA$:Second moment/moment of inertia of cross-section

y : distance to the neutral axis

Pure Bending Normal Stress Formula

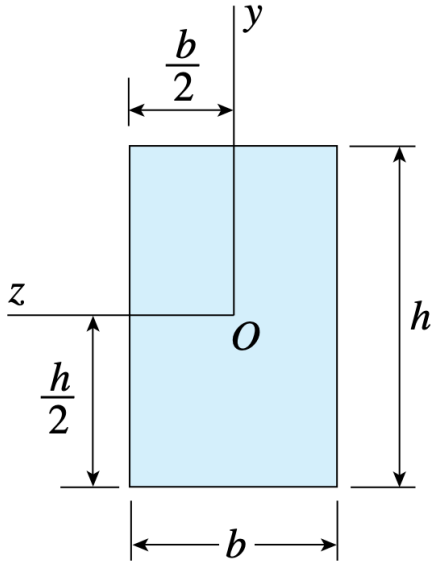
- Normal stress on cross-sections: $\sigma = \dfrac{M_z y}{I_Z}$ (y : distance to the neutral axis)
- Maximum normal stress on cross-sections:

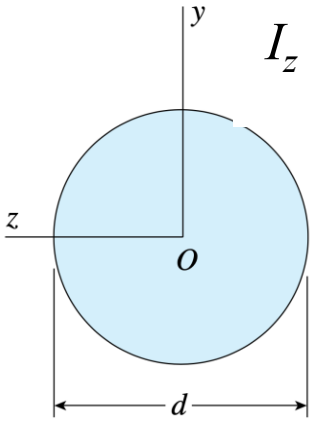
$$\sigma_{max} = \frac{M_{max} y_{max}}{I_Z} = \frac{M_{max}}{W_Z}$$

Bending section modulus, $W_z = I_Z / y_{max}$, represents the **cross-section properties related to bending** in one single quantity

$$I_z = \frac{bh^3}{12} \qquad I_y = \frac{hb^3}{12}$$

$$W_z = \frac{bh^2}{6} = 0.167Ah$$





$$I_z = I_y = \frac{\pi r^4}{4} = \frac{\pi d^4}{64}$$

$$W_z = \frac{\pi d^3}{32} = 0.125Ad$$

Limitations

Normal stress on cross-sections: $\sigma = \frac{M_z y}{I_Z}$

- This flexure formula is derived for pure bending of prismatic beams composed of homogeneous, linearly elastic materials.

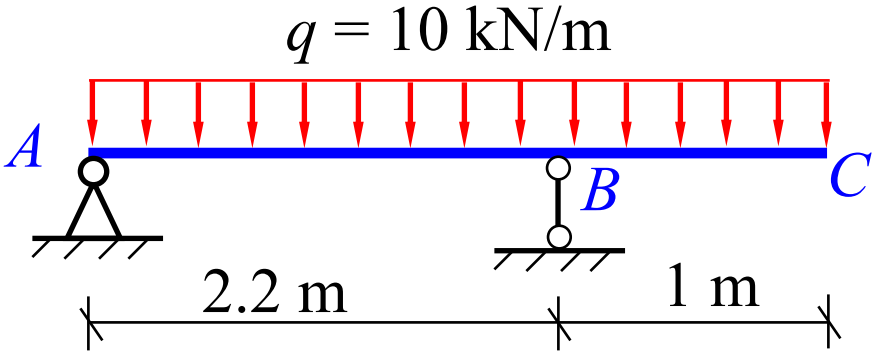
- If a beam is subjected to nonuniform bending, the shear forces will produce warping (or out-of-plane distortion) of the cross sections. **Thus, a cross section that was plane before bending is no longer plane after bending.**

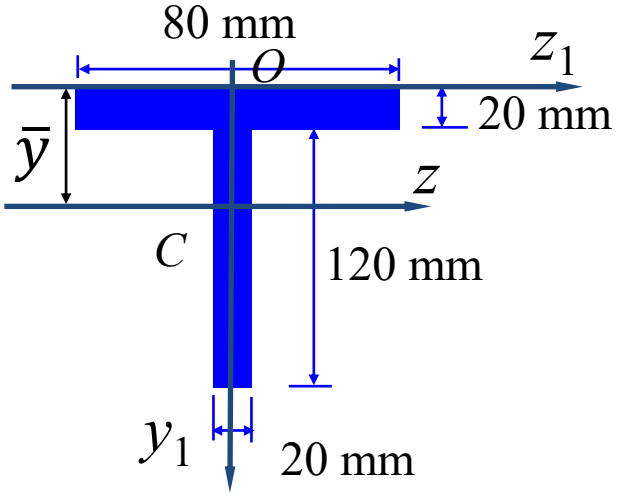
- However, detailed investigations show that the normal stresses calculated from the flexure formula are **not significantly altered** by the presence of shear stresses and the associated warping.

- **Thus, we can use the theory of pure bending for calculating normal stresses in beams subjected to nonuniform bending.**

Example 3

Question: Find the maximum tensile and compressive stress in the *T*-beam shown below.





T-shape Cross Section

Solution:

1. Centroid (neutral surface, neutral axis):

$$\bar{y} = 52 \text{ mm}$$

For details, please refer to Example 2

2. Moment of inertia:

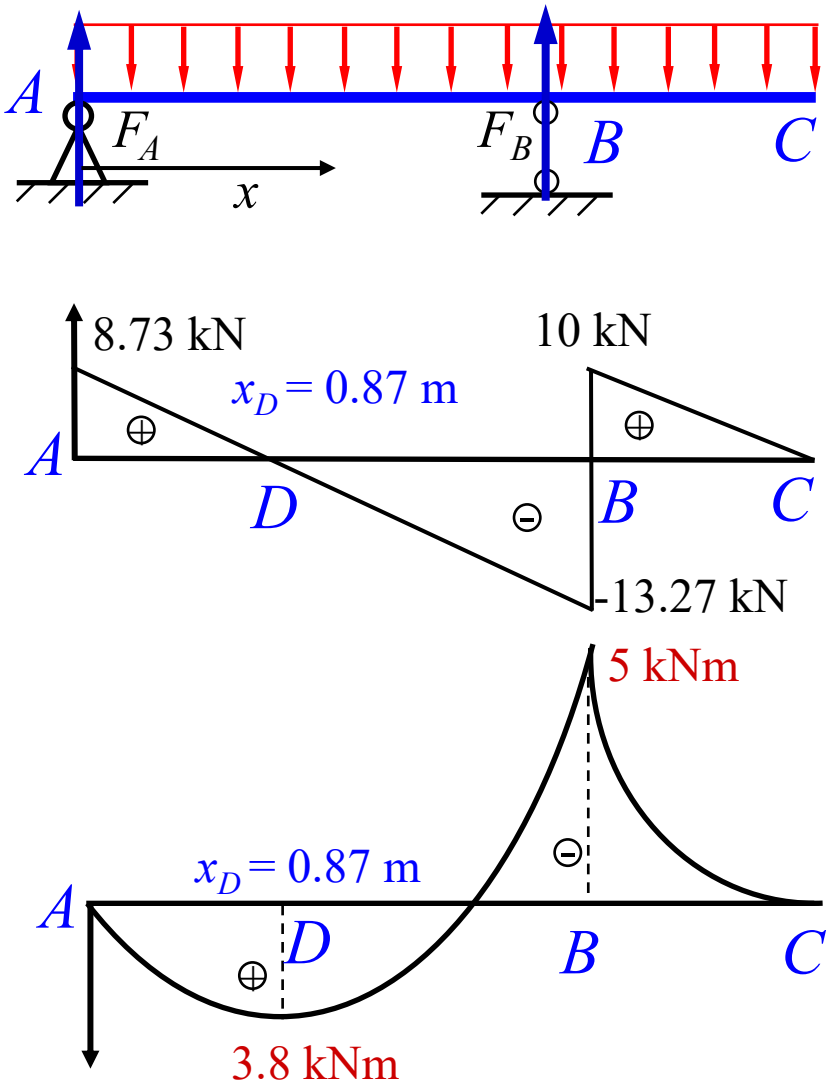
$$I_z = 7.64 \times 10^{-6}\, m^4$$

Example 3

3. Reaction forces and diagram of bending moments

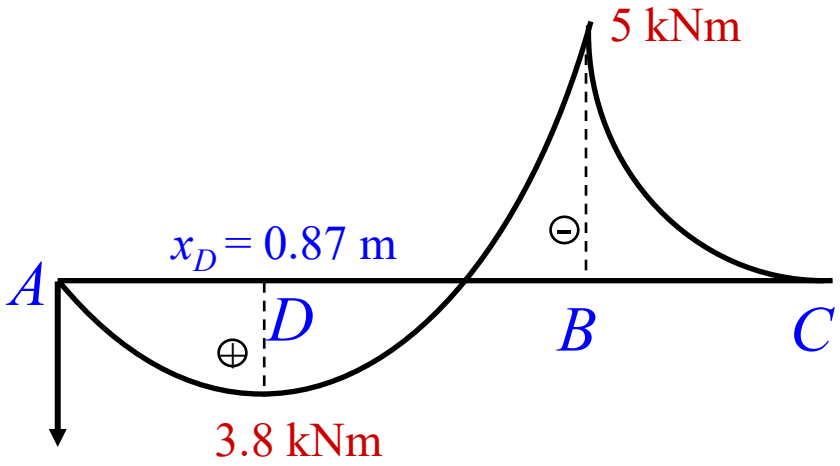
$$0 = \sum M_{Ay} \Rightarrow F_B = 23.27\text{KN}$$

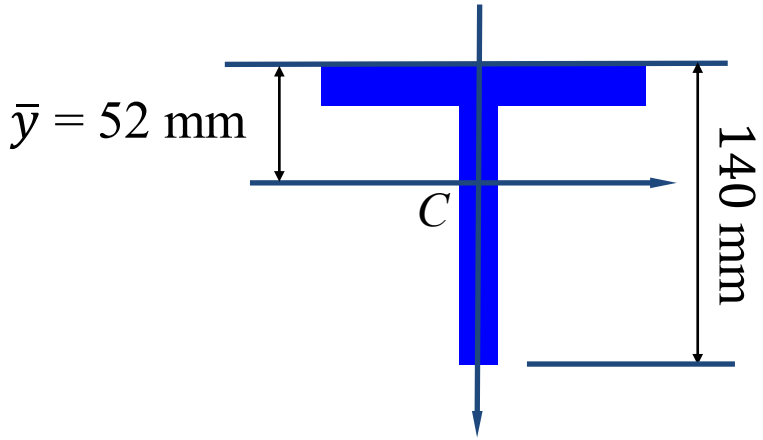
$$\Rightarrow F_A = 10 \times 3.2 - F_B = 8.73\text{KN}$$



Example 3

4. Maximum normal stress (**At cross-section *B***)





Maximum tensile

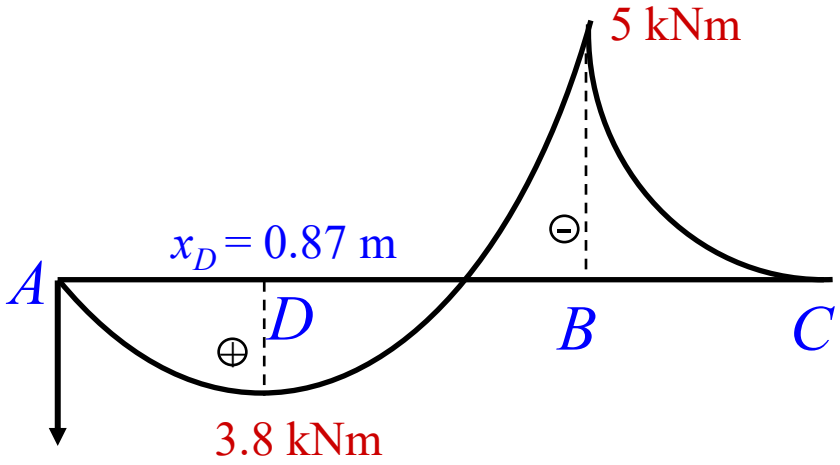
$$\sigma_{\max}^{+} = \frac{(-5\times10^{3})\times(-52\times10^{-3})}{7.64\times10^{-6}} = 34\times10^{6}\ \text{Pa} = 34\ \text{MPa}$$

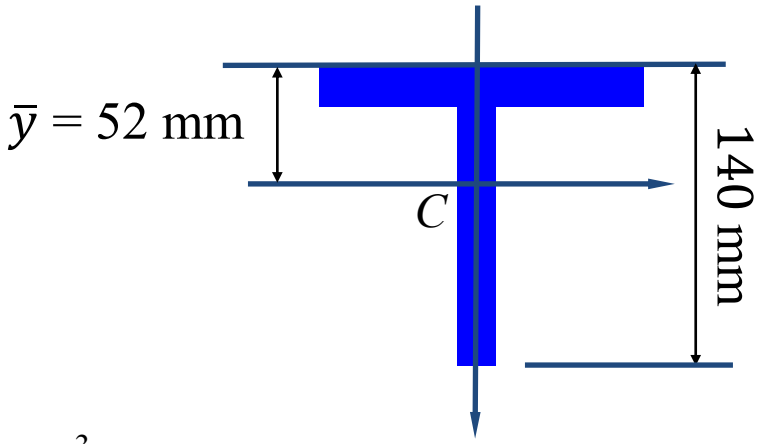
Maximum compressive

$$\sigma_{\max}^{-} = \frac{(-5\times10^{3})\times[(140-52)\times10^{-3}]}{7.64\times10^{-6}} = -57.6\times10^{6}\ \text{Pa} = -57.6\ \text{MPa}$$

Example 3

5. Maximum normal stress (**At cross-section *D***)





Maximum tensile

$$\sigma_{\max}^{+} = \frac{(3.8\times10^{3})\times[(140-52)\times10^{-3}]}{7.64\times10^{-6}} = 43.8\times10^{6}\ \text{Pa} = 43.8\ \text{MPa}$$

Maximum compressive

$$\sigma_{\max}^{-} = \frac{(3.8\times10^{3})\times[-52\times10^{-3}]}{7.64\times10^{-6}} = -25.86\ \text{MPa}$$

6. Maximum normal stress of the beam

- Maximum tensile stress: lower edge of cross-section *D* (43.8 MPa).
- Maximum compressive stress: lower edge of cross-section *B* (-57.6 MPa).

The Position with the Maximum Bending Moments

The maximum positive and negative bending moments in a beam may occur at the following positions:

- A cross section where a concentrated load is applied and the shear force changes sign.
- A cross section where the shear force equals zero.
- A point of support where a vertical reaction is present.
- A cross section where a couple is applied.

Remark: The maximum tensile stress and the maximum compressive stress sometimes don't occur on the same cross-section.

Normal Stress Strength Condition

- Normal stress plays the most important role in satisfying the strength condition of beams under bending.

Normal Stress Strength Condition: $\sigma_{max} = \frac{M_{max}}{W_Z} \leq [\sigma]$ (allowable stress)

- Three types problem that are typically addressed by strength analysis:

  a) Strength check : To determine whether the beam will failure (materials fracture/yielding) $\sigma_{max} = \frac{M_{max}}{W_Z} \leq [\sigma]$

  b) Cross-section design : To select/design the cross-section of beams to minimize materials consumption, for carrying a given load. $W_Z \geq \frac{M_{max}}{[\sigma]}$

  c) Allowable load : To determine the maximum external load that a given beam can carry. $M_{max} \leq [\sigma]W_Z$

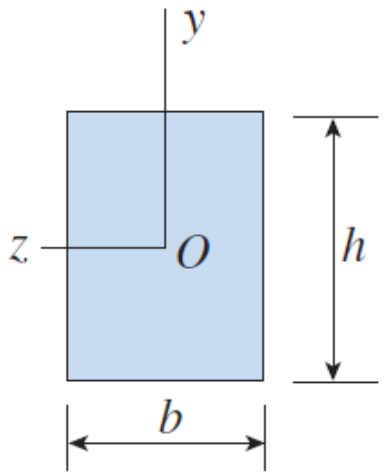
Rational Design of Beams

- When designing a beam to resist bending stresses, we usually begin by calculating the **required bending section modulus, W_z**

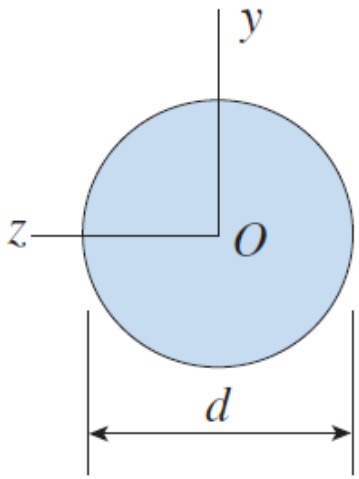
$$W_Z \geq \frac{M_{max}}{[\sigma]}$$

- To **minimize** weight and save materials, we usually select a beam that has the **least cross-sectional area** while still providing the required section modulus

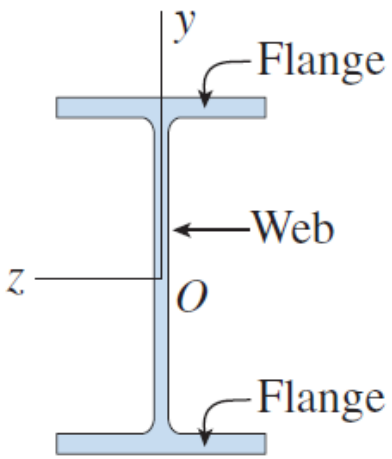
Rational Design of Cross-sections



$$W_z = \frac{bh^2}{6} = 0.167Ah$$



$$W_z = \frac{\pi d^3}{32} = 0.125Ad$$



$$W_z \approx 0.35Ah$$

Note: A = cross-sectional area

- The most efficient beam is one in which the material is located as far as practical from the neutral axis.
- The farther a material is from the neutral axis, the larger the section modulus becomes, and the larger the bending moment that can be resisted.

I-Shape Cross-sections in Beams







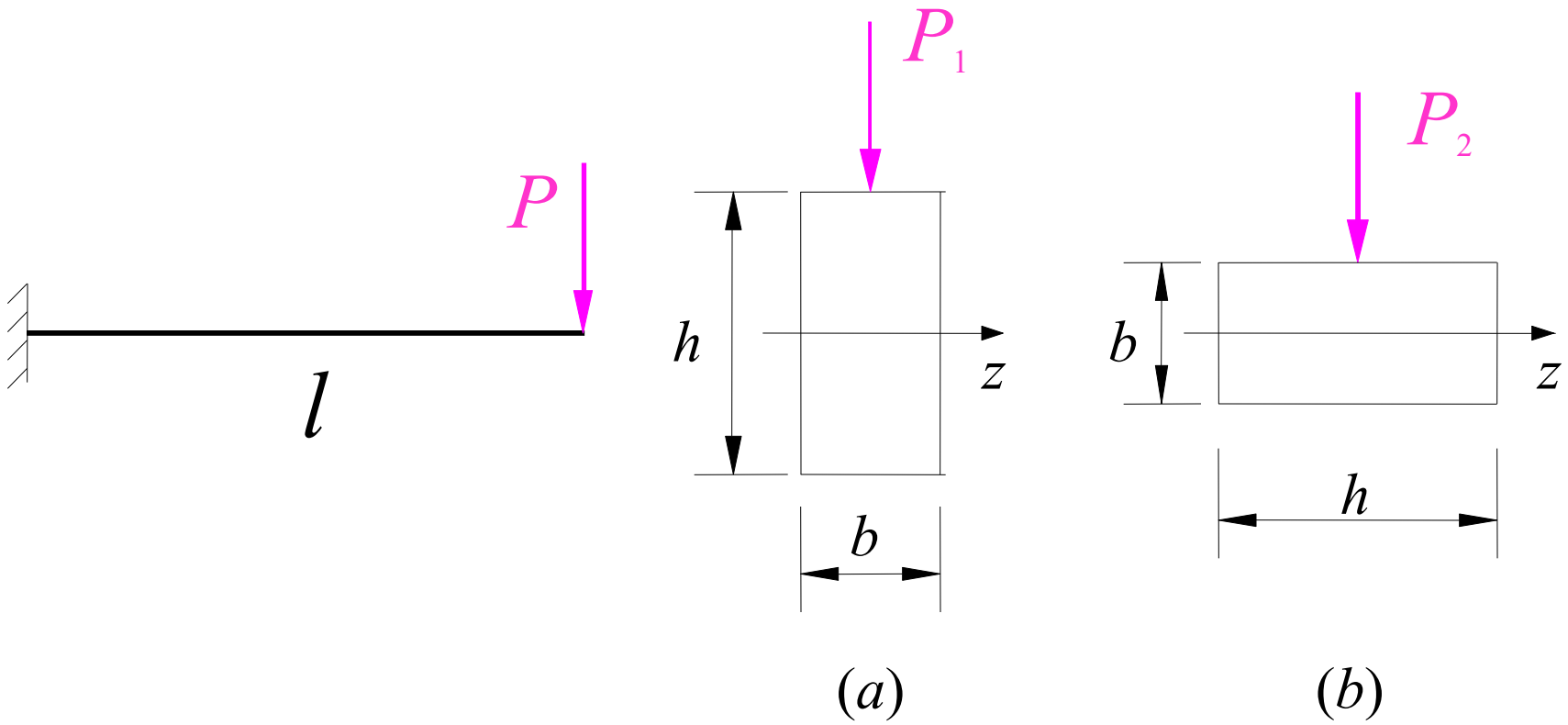






Example 4

- The dimension and material of the two cantilever beams shown are identical. Find the allowable load ratio of these two beam based on the normal stress strength condition: $P_1/P_2 = ?$



Example 4

- Solution：

$$\sigma_{\max 1}=\frac{M_{\max 1}}{W_{z1}}=\frac{P_1 l}{\dfrac{bh^3}{12}\Big/\dfrac{h}{2}}=\frac{P_1 l}{\dfrac{bh^2}{6}}$$

$$\sigma_{\max 2}=\frac{M_{\max 2}}{W_{z2}}=\frac{P_2 l}{\dfrac{hb^2}{6}}$$

$$\sigma_{\max 1}=\sigma_{\max 2}\Rightarrow\frac{P_1}{P_2}=\frac{h}{b}$$

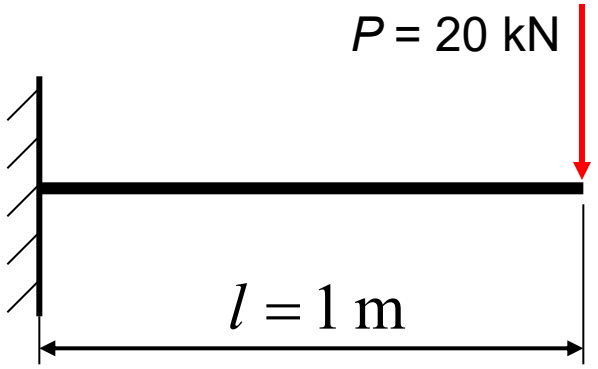
Example 5

- Given $P = 20$ kN, $[\sigma] = 140$ MPa. Compare the material consumption for the following three types of cross-sections:

(1) rectangle with $h/b = 2$; (2) circle; (3) I-shaped.

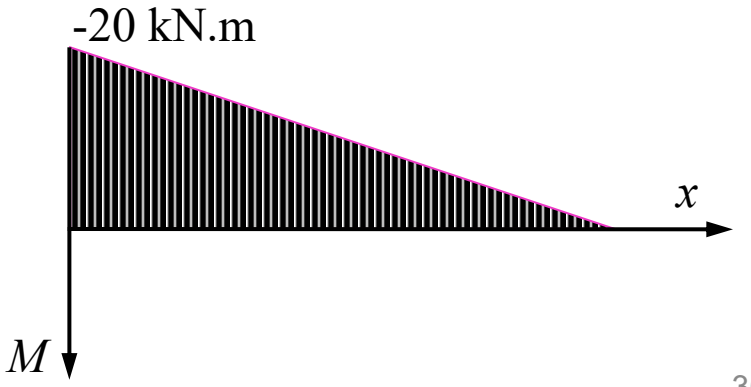
- Solution:

Diagram of bending moment.



$$\sigma_{\max} = \frac{M_{\max}}{W_z} \le [\sigma] \Rightarrow$$

$$W_z \ge \frac{M_{\max}}{[\sigma]} = \frac{20 \times 10^3}{140 \times 10^6}\ \mathrm{m}^3 \approx 143\ \mathrm{cm}^3$$



(1) For rectangular cross-section

$$W_z = \frac{bh^2}{6} \Rightarrow A_1 = 72\ \mathrm{cm}^2$$

Example 5

(2) For circular cross-section

$$W_z = \frac{\pi d^3}{32} \Rightarrow d = 11.3\ \text{cm} \quad \Rightarrow \quad A_2 \approx 100\ \text{cm}^2$$

(3) For *I*-shaped cross-section

Check the table for I-beam:

I-beam consumes the least material while circular beam costs the most.

$$W_z = 144.1\ \text{cm}^3 \quad \Longrightarrow \quad A_3 = 34.01\ \text{cm}^2$$

Designation	Mass per meter	Area of section	Depth of section	Width of section	Thickness		Strong axis 1-1		
	G	A	h	b	t_w	t_f	I_1	S_1	r_1
	kg/m	cm^2	mm	mm	mm	mm	cm^4	cm^3	cm
HE 180 B	51.2	65.25	180	180	8.5	14	3831	425.7	7.66
HE 160 B	42.6	54.25	160	160	8	13	2492	311.5	6.78
HE 140 B	33.7	42.96	140	140	7	12	1509	215.6	5.93
HE 120 B	26.7	34.01	120	120	6.5	11	864.4	144.1	5.04
HE 140 A	24.7	31.42	133	140	5.5	8.5	1033	155.4	5.73

Area of section: A Bending section modulus: W_z

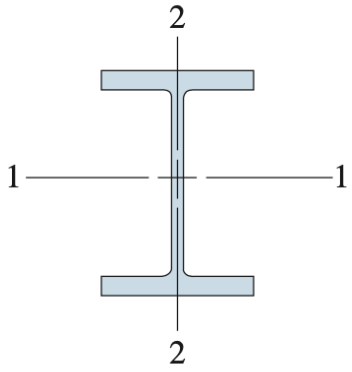


Table is available in Appendix E of book "**Mechanics of Materials, Eighth Edition, SI** by James M. Gere and Barry J. Goodno "

THE END